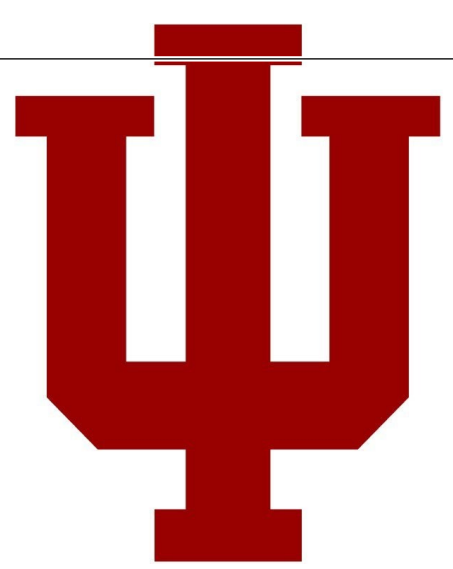




Global Controls on Seasonal Forecast Skill in the Tropics

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Objectives

Unlike machine learning (ML) weather prediction models that surpass traditional numerical weather models in many global metrics, ML climate simulations or seasonal outlook are still limited. This is because long-term ML model rollout approaches the climatology of training data after the model reaches its limited predictability range. In this study, we explore the use of a data-driven architecture, so-called Reservoir Computing (RC) to examine the large-scale variability of the tropical atmosphere. Our specific aims are 1) explore the RC's ability for ML-based climate seasonal outlook and 2) quantify the dependence of tropical variability on data channels and global environments.

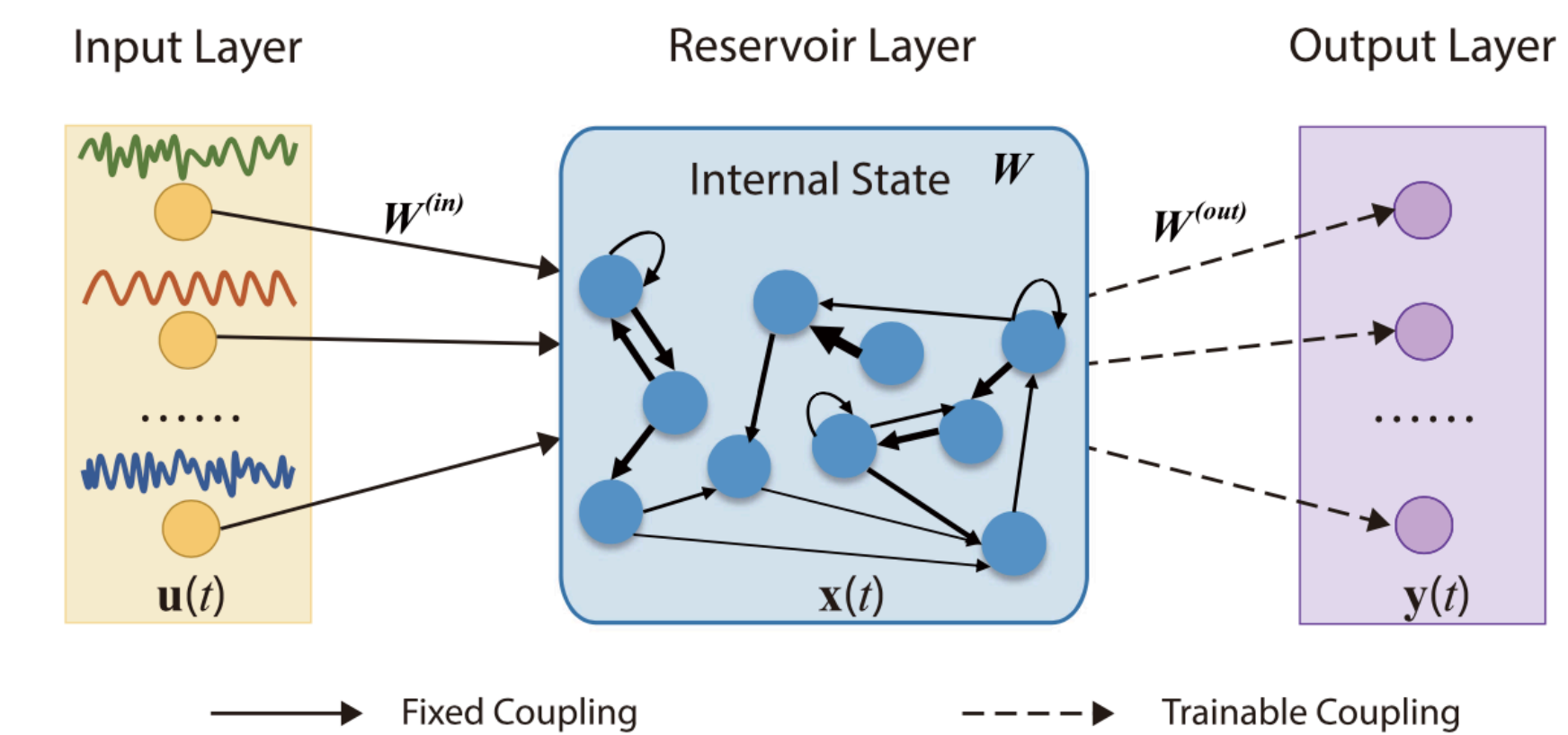


Figure 1. An architecture of an RC model with an input layer, the reservoir, and the output layer, which is designed for chaotic dynamical systems (Yan et al. 2024)

Model Design & Data

- An RC model consisting of 3 components based on the PyRCN package was designed (Fig. 1), which was verified against a chaotic system (Lorenz 1963) with ridge regression for training the output layer;
- The NCEP reanalysis (1948-2020) was used to decompose global atmospheric states into principal components to reduce the data dimension, which are treated as input features for the RC model;
- The RC model was trained with 67 years, with the last 5 years saved for testing;
- Model variables include sea level pressure, zonal wind (850 hPa), temperature (750 hPa), and relative humidity (500 hPa);

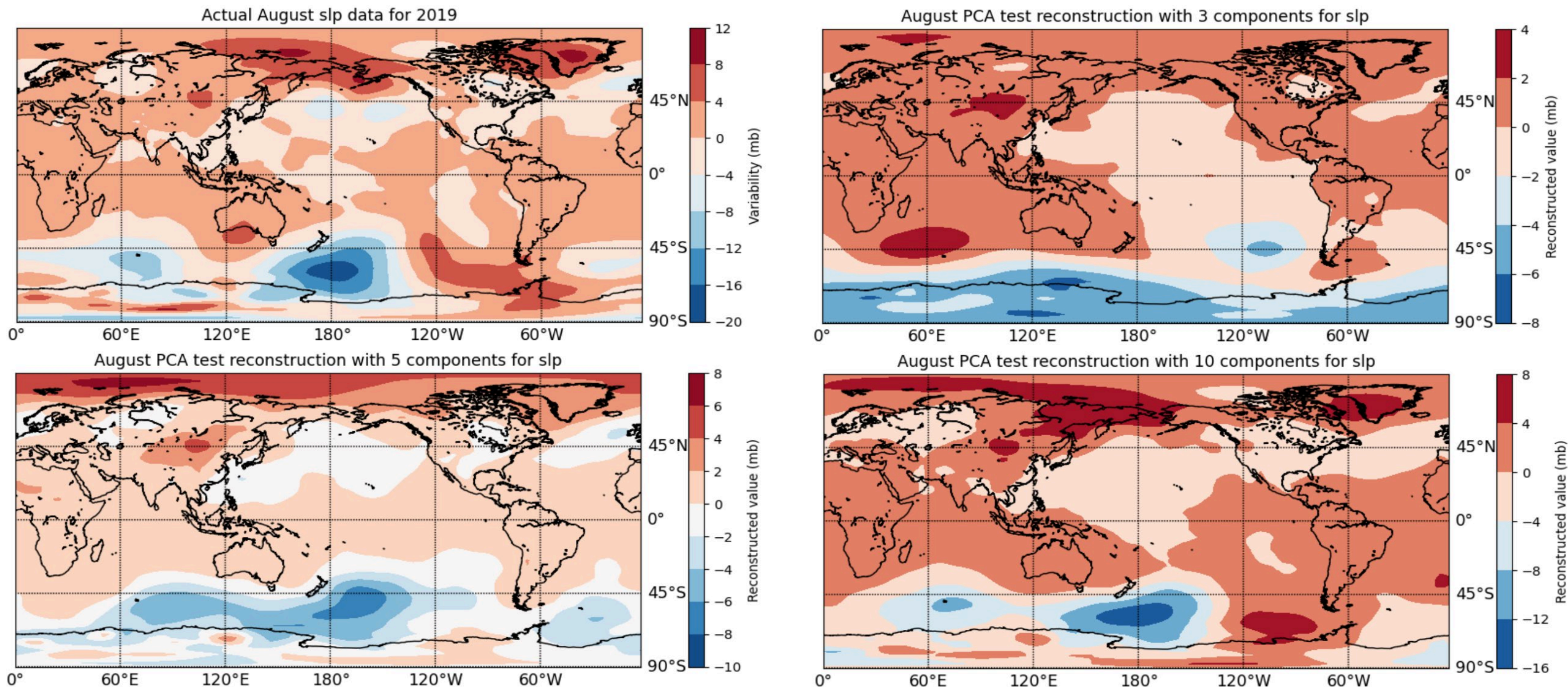


Figure 2. Principal component analyses (PCA) of sea level pressure anomaly for August 2019. Actual anomaly (Top Left), reconstruction of anomaly using 3 components (Top Right), 5 components (Bottom Left), and 10 components (Bottom Right).

Result # 1: PCA-dimension reduction

- Using more PCA components allows the RC model to capture more of the trends seen in the original data (Fig. 3), with forecast skill decaying after 3-month lead time;
- The predictability of the SLP anomaly decreases as the number of components used in PCA increases. This may be the result of additional components acting as noise (Fig. 3);
- A linear relationship between the test and predicted reconstructions for 1 variable exists, producing a correlation coefficient of ± 1 , but the reconstructions still differ.

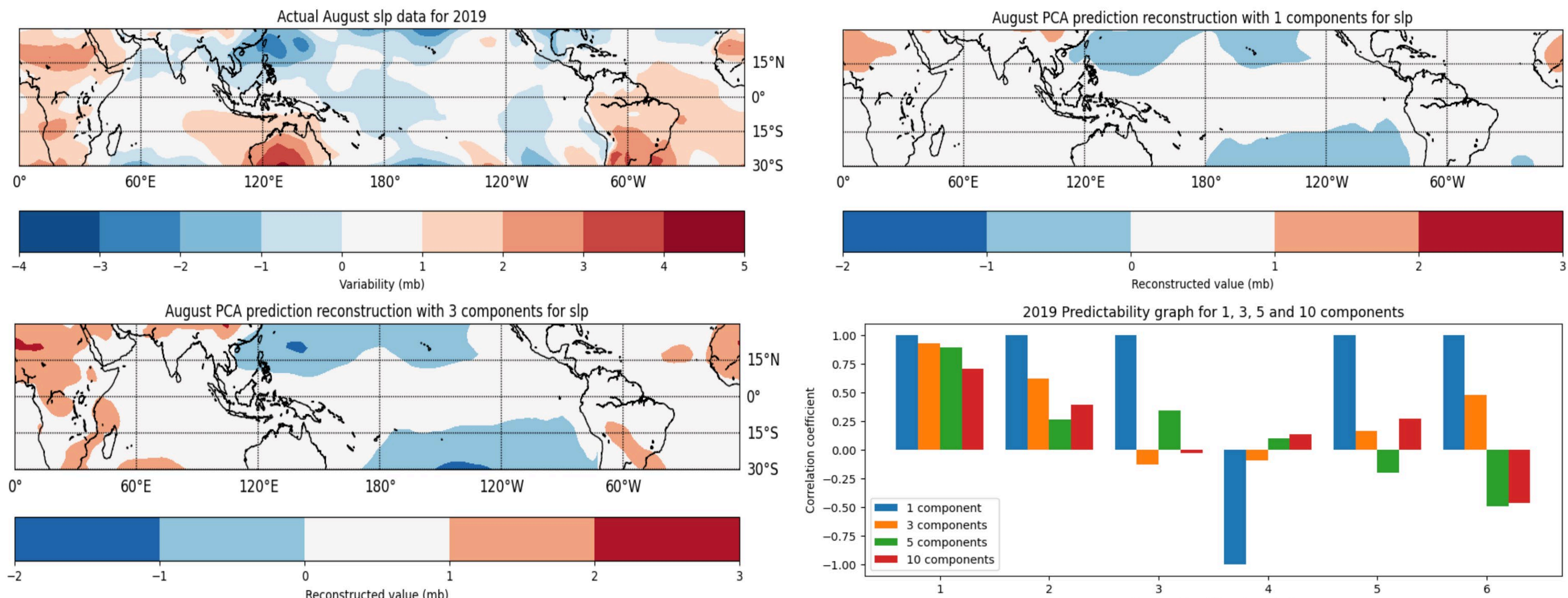


Figure 3. Predicting the tropical Sea Level Pressure anomaly for one test sample (August 2019). Actual anomaly (Top Left), reconstruction of anomaly using prediction for 1 components (Top Right) and 3 components (Bottom Left). Correlation time series as a function for forecast lead time for 1, 3, 5 and 10 components (Bottom Right).

Result # 2: Global teleconnection

- Predictability of SLP generally decays after the first few months, but it recovers when using 3 PCA components (Fig. 3-4);
- Using 3 components, the domain within a latitude range of 60 S to 60 N shows the most consistent predictability over a 6-month lead time when using only SLP (Fig. 4);
- Predictability for a global domain tends to be higher for 5 components than for 3 components (Fig. 4), suggesting that large-scale predictability in the tropical region requires global information instead of just inside the tropical region;

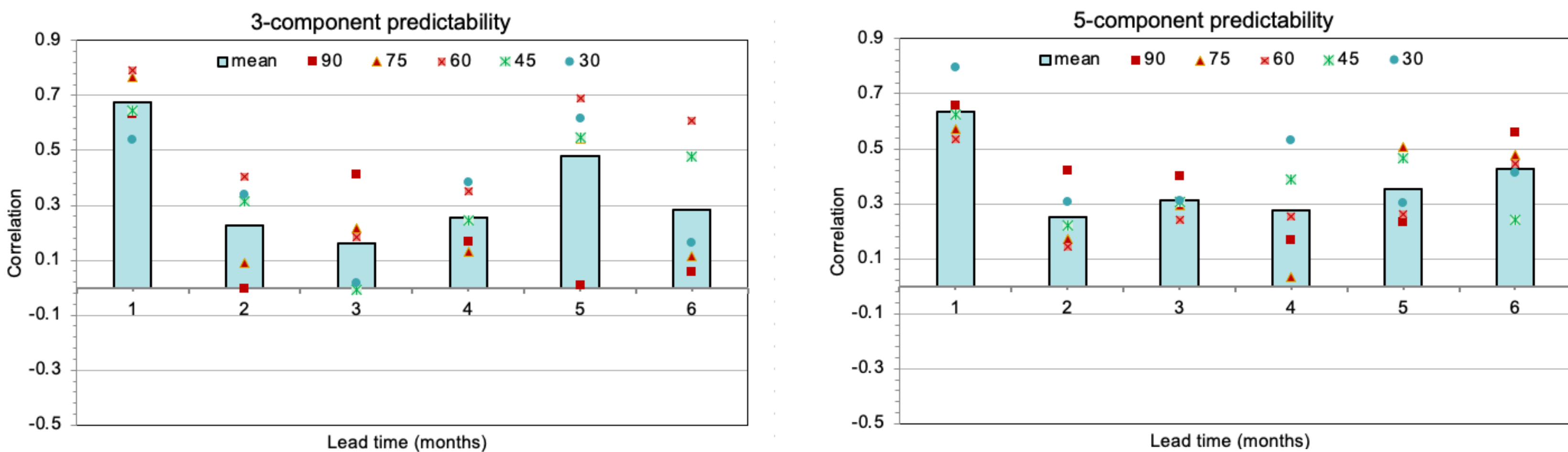


Figure 4. Mean correlation between the ground truth and RC model prediction during the test period 2016-2020 for a ranges of domain latitude, using May-July input to predict future components.

Result # 3: Cross-Variable Effects

- Using additional variables to predict SLP shows degraded predictability in the tropical region (30S-30N) at 1-3-month lead time, when using both 3 and 5 components (Fig. 5);
- The impact of additional features on predictability depends on domain size and lead time. Specifically, using 5 PCA components improved the overall predictability for a 2-6 month lead time, but degraded predictability for the 3-component experiments at all lead times (Fig 4-5);
- Feature engineering by stratifying input features according to month, variable, and component number did not significantly change predictability.

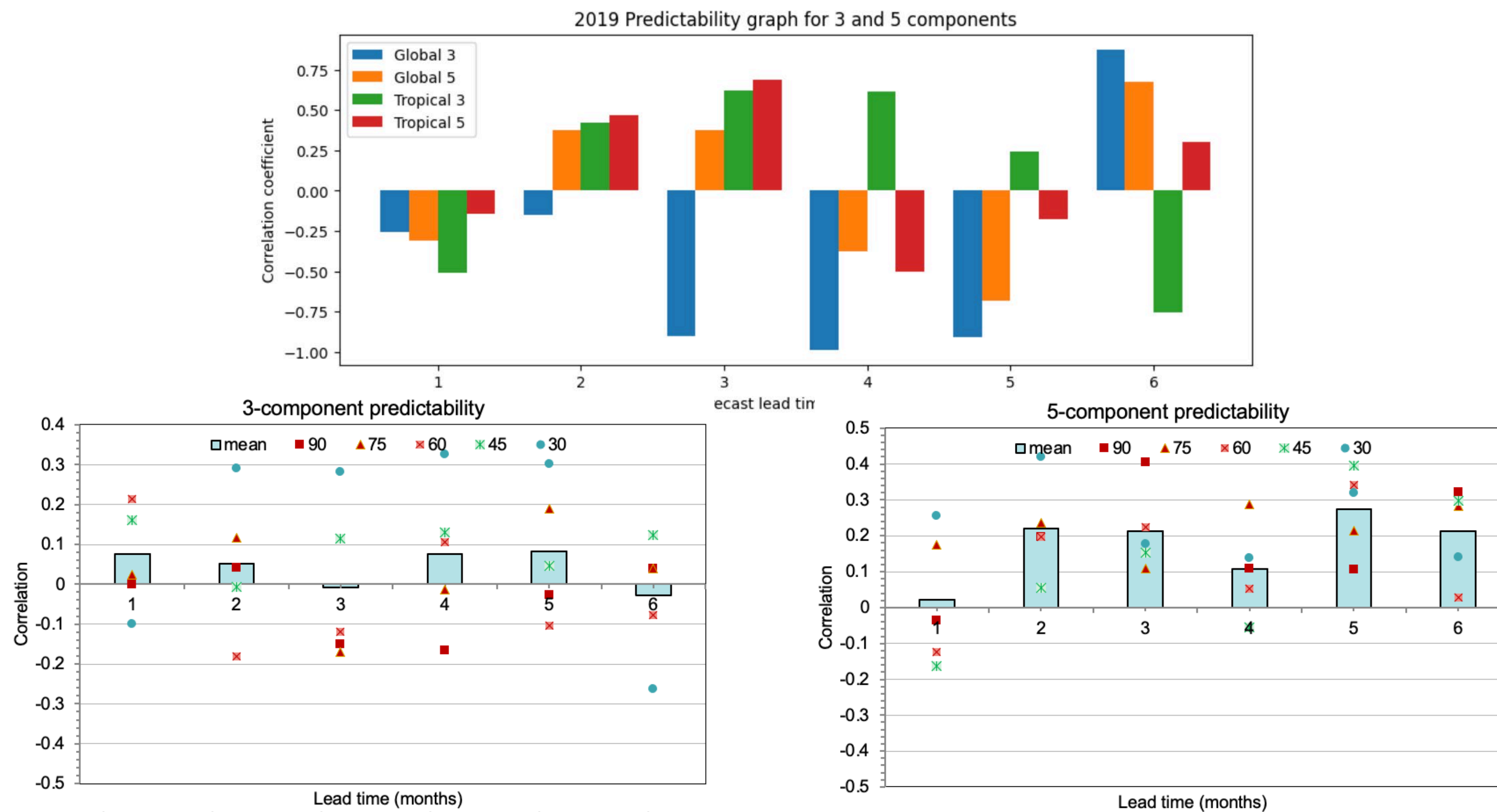


Figure 5. Mean correlation between RC prediction and ground truth with additional variables for May-July using 3 components (Bottom Left) and 5 components (Bottom Right). Correlation for a test year of 2019 with global and tropical (30S/N) domain with the same training data (Top).

Conclusions

- Using PCA to decompose and reduce the dimension of the large-scale variability can improve prediction of large-scale states up to 3-month lead time;
- RC models are promising to handle highly-varying atmospheric states in a reduced-dimension phase space, which captures a consistent decay of predictability before recovering at longer lead times of 4-5 months;
- Including global information does not seem to help improve the overall predictability of tropical variability, suggesting the de-coupling of large-scale dynamics in the tropical region for seasonal forecast;
- Predictability depends on PCA components, domain size, and input features. More work to be done to establish an exact relationship among these factors;

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